The empirical recurrence for OEIS A297680: recovering a conjecture that is not written down

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Abstract

OEIS A297680 records "Empirical recurrence of order 57", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 152$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 57, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 57 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A297680 is "Number of nX6 0..1 arrays with every 1 horizontally, diagonally or antidiagonally adjacent to 0, 1 or 4 neighboring 1s.". It has offset 1 and begins

$$44, 597, 12110, 221420, 4202440, 78957968, 1487819051, 28013761161, \ \ldots$$

The entry states:

> Empirical recurrence of order 57 (see link above)

As of the "Last modified" line on the live entry (Jan 03 2018 09:27:39, revision 4) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 4096$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 152$ of the 4096, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 152$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 304$ exact terms of the model returns order 57 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{57} c_i \, a(n-i),$$

c_1	15	c_{16}	−870179290	c_{31}	10995238771	c_{46}	−29087943926
c_2	64	c_{17}	−2018857923	c_{32}	−508149300865	c_{47}	38224881476
c_3	159	c_{18}	629759167	c_{33}	−385381195512	c_{48}	19922318002
c_4	226	c_{19}	1761693438	c_{34}	340089661655	c_{49}	−9912057436
c_5	−6649	c_{20}	11822456383	c_{35}	565590918462	c_{50}	−17667626192
c_6	19028	c_{21}	−4718396914	c_{36}	−228153340847	c_{51}	−4701957008
c_7	−131845	c_{22}	−29776319229	c_{37}	−525384680638	c_{52}	5468201720
c_8	−651867	c_{23}	10073922941	c_{38}	−440764935195	c_{53}	1640871312
c_9	1167620	c_{24}	23274737849	c_{39}	100303121724	c_{54}	−770923776
c_{10}	13904108	c_{25}	39561341512	c_{40}	375145831447	c_{55}	−102469696
c_{11}	−9855884	c_{26}	−10698993171	c_{41}	−32512630854	c_{56}	37042656
c_{12}	−82653737	c_{27}	−217202926268	c_{42}	−258604888548	c_{57}	−7168896
c_{13}	−4507473	c_{28}	−40100409884	c_{43}	30194427733		
c_{14}	220678729	c_{29}	320953354764	c_{44}	142189419597		
c_{15}	593758182	c_{30}	244199486044	c_{45}	−2105332180		

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 57, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $57 + 4$ terms removed returns order 57 as well, so $\deg q_{\min} = 57 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $57 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 57 that this sequence satisfies — and that family turns out to have exactly one member.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A297680.

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